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COMMON QUARTERLY EXAMINATION - 2026

A Standard XII Reg.No. | | | | | | |

CHEMISTRY

Time : 3.00 hrs Part – I *Marks : 70*

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I. Choose the correct answer. 15 x 1 = 15

1. Flux is a substance which is used to convert
   a) Mineral into silicate b) Infusible impurities into soluble impurities
   c) Soluble impurities into infusible impurities d) All of these
2. Number of five-membered rings present in Buckminster fullerene is.
   a) 12 b) 20 c) 50 d) 60
3. The compound that is used in nuclear reactors as protective shields and control rods is:
   a) Metal borides b) Metal oxides c) Metal carbonate d) Metal carbides
4. Among the following, which is the strongest oxidizing agent?
   a) Cl_2 b) F_2 c) Br_2 d) I_2
5. Salts of nitric acid are used in gunpowder for firearms. Which of the following is used?
   a) $AgNO_3$ b) KNO_3 c) $NaNO_3$ d) NH_4NO_3
6. Permanganate ion changes to _____ in acidic medium.
   a) MnO_4^{2-} b) Mn^{2+} c) Mn^{3+} d) MnO_2
7. Solid CO_2 is an example of:
   a) Covalent Solid b) Metallic Solid c) Molecular Solid d) Ionic Solid
8. Assertion: Rate of reaction doubles when the concentration of the reactant is doubled if it is a first-order reaction.
   Reason : Rate constant also doubles.
   a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
   b) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
   c) Assertion is true but Reason is false d) Both Assertion and Reason are false
9. The addition of a catalyst during a chemical reaction alters which of the following quantities?
   a) Enthalpy b) Activation energy c) Entropy d) Internal energy
10. Which of the following salt is acidic in nature?
    a) $NaNO_3$ b) CH_3COONa c) NH_4Cl d) Both (b) and (c)
11. The pH of 10^{-5} M KOH solution will be
    a) 9 b) 5 c) 19 d) None of these
12. On reacting with neutral ferric chloride, phenol gives:
    a) Red colour b) Violet colour c) Dark green colour d) No colouration
13. Iso propyl benzene, on air oxidation in the presence of dilute acid, gives
    a) C_6H_5COOH b) $C_6H_5COCH_3$ c) $C_6H_5COC_6H_5$ d) C_6H_5OH
14. The reagent used to distinguish between acetaldehyde and benzaldehyde is
    a) Tollens reagent b) Fehling's solution
    c) 2,4-dinitrophenyl hydrazine d) Semi carbazide
15. _____ is used in the manufacture of thermosoftening plastic Perspex?
    a) Benzaldehyde b) Acetone c) Acetaldehyde d) Benzophenone

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Part - II

II. Answer any six questions. (Q.No.24 is compulsory) 6 x 2 = 12

16. What is the role of quick lime in the extraction of Iron from its Fe_2O_3?
17. Write the uses of di borane.
18. What happens when PCl_5 is heated?
19. Which metals present in 3d series exhibit +1 oxidation state most frequently and why?
20. What is Packing efficiency?
21. Give two examples for zero order reaction.
22. Write the limitations of Arrhenius theory.
23. Write Schotten–Baumann reaction.
24. Find out compound A and B in the following reactions.

$$CH_3COOH \xrightarrow{SOCl_2} A \xrightarrow{LiAlH_4} B$$

Part - III

III. Answer any six questions. (Q.No.33 is compulsory) 6 x 3 = 18

25. Write the electronic configuration of Ce^{4+} and Co^{2+}.
26. Give the basic requirement for Vapour–Phase method.
27. Draw the structure of CO and CO_2.
28. Write a note on Holme's Signal.
29. Explain Pseudo first order reaction with an example.
30. Write the difference between Lewis acid and Lewis base.
31. Write the Swern oxidation reaction.
32. What is Rosenmund Reduction? What is the purpose of adding $BaSO_4$ in it?
33. Calculate the expression for the solubility product of Hg_2Cl_2.

Part - IV

IV. Answer all the questions. 5 x 5 = 25

34. a) Explain zone refining process with an example. *(OR)*
    b) i) Describe the structure of Diborane.
    ii) How will you convert boric acid into boron nitride?
35. a) i) Explain the test for sulphate with suitable chemical equation.
    ii) Complete the following reactions :
    a) $AgNO_3 + PH_3 \longrightarrow$ b) $Xe + F_2 \xrightarrow[400°C]{Ni\ /\ 200\ atm}$ *(OR)*
    b) Compare Lanthanoids and Actinoids.
36. a) i) Explain Frenkel defect. ii) Write Bragg's equation and explain its terms.
    (OR)
    b) i) Differentiate between order of reaction and molecularity.
    ii) Write the rate law for the reaction in which the order with respect to reactants x, y, and z are 1/2, 3/2, and zero respectively.
37. a) Derive Henderson equation for acid buffer solution. *(OR)*
    b) Write short notes on the following: (i) Reimer–Tiemann reaction
    (ii) Knoevenagel reaction (iii) Benzoin Condensation
38. a) i) How does diethyl ether react with the following:
    a) Cl_2 / light b) dil H_2SO_4 c) PCl_5
    ii) Give coupling reaction of phenol. *(OR)*
    b) Explain the mechanism of Cannizzaro reaction.

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